

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech. Branch : Information Technology Semester : VII

Subject Code & Name: Machine Learning (BTITOE704B)

Max Marks: 60

Date: 07/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) Describe in detail the steps involved in designing a learning system. **L1 6**
- B) Define learning problem. Describe the following problems with respect to task, performance and experience: **L2 6**
- 1) Handwritten Recognition problem.
- 2) Checkers learning problem.
- 3) Fruit Prediction problem.
- C) Find the specific and general hypothesis using candidate elimination algorithm for following. **L3 6**

Sky	Temperature	Humid	Wind	Water	Forest	Output
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
rainy	cold	high	strong	warm	change	no
sunny	warm	high	strong	cool	change	yes

Q.2 Solve Any Two of the following.

- A) Describe Hypothesis testing in ML. **L2 6**
- B) Write a short note on sampling theory. **L2 6**
- C) Define entropy and information gain and describe steps of ID3 Algorithm. **L2 6**

Q. 3 Solve Any One of the following.

- A) Explain Bayes theorem in detail. **L3 6**
- B) Write a short note on PAC learnability. **L3 6**

- C) Classify the following instance using Naive Bayes classifier:
Instance = (Red, SUV, Domestic).

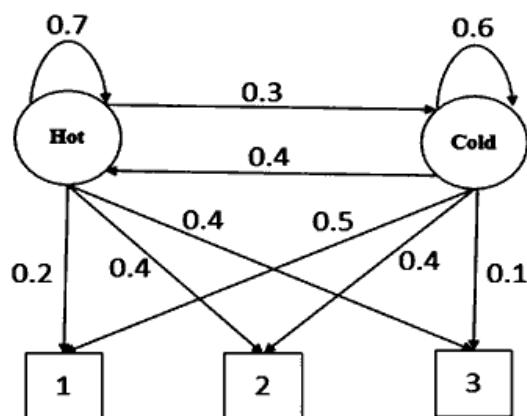
L4

6

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Q.4 Solve Any Two of the following.

- A) Demonstrate Bias-variance decomposition with diagram L3 6
- B) Explain SVM in detail. L1 6
- C) Find the hidden state sequence for the observable sequence [3, 1, 3]. L4 6
where $\pi = [0.8 \text{ (Hot)}, 0.2 \text{ (Cold)}]$



Q.5 Solve Any One of the following.

- A) Explain HMM in detail L2 6
- B) Write a short note on Q-learning L2 6
- C) Write a short note on non-deterministic rewards and action L2 6

*** End ***